UNIT 4 · STANDARD 6.RP.3C

Use Percent to Solve Problems

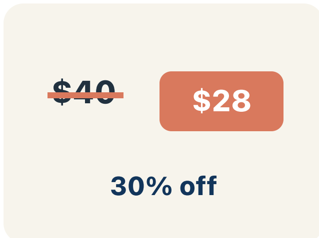
Lesson 4-5

Name: ____________________ Date: ____________________ Class: ____________________

Key Vocabulary

Level 1 support

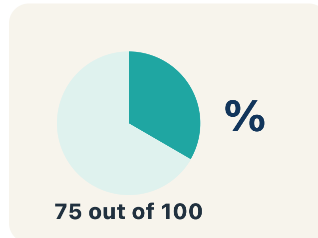
Picture first, then the word, then a plain-language meaning. Say each word out loud.



20% off $50 → discount is $10

Discount

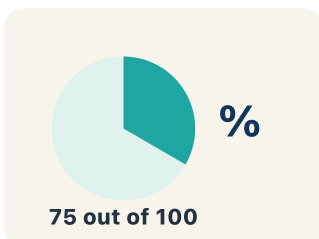
Money taken off the first price to make it cheaper.



A $30 item with 10% markup sells for $33

Markup

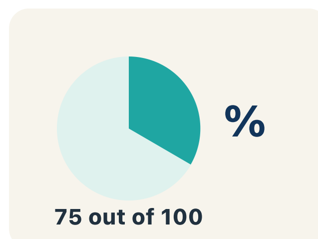
Money added to the cost to set the selling price.



6% tax on $20 adds $1.20

Tax

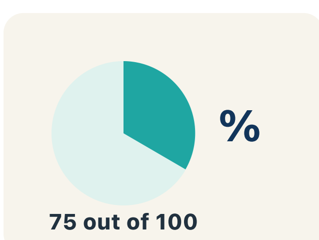
Extra money added to a price for the government.



15% tip on $40 = $6

Tip

Extra money you give for good service.



50% means 50 out of 100

Percent

A way to compare a number to 100, shown with the % sign.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can solve problems involving discounts, markups, tax, and tips using percents.

1 Notice

Welcome to the arcade prize shop!

2 Key idea

I can solve problems involving discounts, markups, tax, and tips using percents.

3 Words I'll use

Discount **Markup** **Tax** **Tip** **Percent**

🤔 Think About It

- What happens to the price when there is a discount?
- What happens to the price when tax is added?
- Does it matter whether you apply the discount or tax first?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: A prize costs 120 tickets and is 25% off. What is the sale price?

A. 90 tickets

B. 95 tickets

C. 100 tickets

D. 30 tickets

Step 1 25% of 120 = 0.25 × 120 = 30.

Step 2 Sale price: 120 − 30 = 90 tickets.

✅ **Answer:** A. 90 tickets

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: A meal costs $20. You leave a 15% tip. How much is the tip?

A. $3.00

B. $1.50

C. $5.00

D. $0.15

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: A shirt costs $40 and the sales tax is 5%. What is the total cost including tax?

A. $42.00

B. $45.00

C. $40.05

D. $38.00

Show your work:

Turn & Talk — Discussion Points (Level 1 support) (Level 2)

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

At the prize shop a bear is 25% off but the shop adds 5% tax. Which of these raises the price and which lowers it, and what do the words discount, markup, tax, and tip have in common?

Level 1 support

Start here: A discount lowers the price, while tax, tip, and markup raise it.

💡 **Need a hint?**

Sentence starters (optional)

A ____ makes the price go down, but a ____ makes it go up.

Un ____ hace que el precio baje, pero un ____ hace que suba.

They are alike because each is a ____ of the price.

Se parecen porque cada uno es un ____ del precio.

Word bank: discount | markup | tax | tip | percent

Listen for: *Students sort discount as lowering price and tax/tip/markup as raising it, and recognize all are a percent of a price.*

Level 2

Does it matter whether the shop takes the discount first or adds the tax first? Predict, and explain your reasoning before you calculate.

- I think the order does / does not matter because ____.
- Applying the discount first means ____.

EXPLORE

In your table you found the discount, then the tax. Why do we calculate the tax on the price AFTER the discount instead of the original price?

Level 1 support

Start here: Tax is charged on the lower price you actually pay after the discount.

💡 **Need a hint?**

Sentence starters (optional)

I find the discount first because ___.

Primero hallo el descuento porque ___.

Then I take 5% tax on the ___ price, which is ___.

Luego tomo 5% de impuesto sobre el precio ___, que es ___.

Word bank: discount | tax | percent | markup

Listen for: *Students explain the discount lowers the price first, then tax is applied to that new lower price (e.g., 5% of 72 = 3.6).*

Level 2

A classmate added the 5% tax and the 25% discount into one '20% off' step. Explain why combining them like that gives the wrong final price.

- Combining them is wrong because ___.
- Tax and discount cannot just be added because ___.

CONNECT

An arcade membership is $50, marked 30% off, then 6% tax is added. Your friend says the total is $37.10. Is that right, and how would you prove it?

Level 1 support

Start here: After 30% off the price is $35, then 6% tax makes the total $37.10.

💡 **Need a hint?**

Sentence starters (optional)

First the discount: 30% of $50 = $___, so the sale price is $___.

Primero el descuento: 30% de $50 = $___, así que el precio rebajado es $___.

Then 6% tax on $___ adds $___, for a total of $___.

Luego 6% de impuesto sobre $___ suma $___, para un total de $___.

Word bank: discount | tax | percent | tip

Listen for: *Students compute $15 discount, $35 sale price, $2.10 tax, and $37.10 total, confirming the friend is correct with shown steps.*

Level 2

Suppose the membership were a tip situation instead: $50 plus a 20% tip. Compare that total to the $37.10 discount-then-tax total and explain why one is so much higher.

- A 20% tip makes the total $___ because ___.
- It is higher than $37.10 because ___.

Write About the Math

The Writing Revolution

I can explain my solution using the words discount, markup, tax, tip, and percent.

1. Kernel Sentence

subject + verb

Model: Percent is a way to compare a number to 100, shown with the % sign.
Porcentaje es una manera de comparar un número con 100, con el signo %.

Write a kernel sentence about percent. Use a subject and a verb.
Escribe una oración base sobre porcentaje. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Percent matters in math
Porcentaje importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Percent matters in math because ____.** *Porcentaje importa en matemáticas porque ____.* ______________________________
but *pero*	**Percent matters in math, but ____.** *Porcentaje importa en matemáticas, pero ____.* ______________________________
so *entonces*	**Percent matters in math, so ____.** *Porcentaje importa en matemáticas, entonces ____.* ______________________________

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about percent.
Di un hecho verdadero sobre percent.

Percent ____.

Question
Pregunta

Ask a question about percent.
Haz una pregunta sobre percent.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about percent.
Muestra entusiasmo sobre percent.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with percent.
Dile a un compañero qué hacer con percent.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

A ____ makes the price go down, but a ____ makes it go up.
Un ____ hace que el precio baje, pero un ____ hace que suba.

They are alike because each is a ____ of the price.
Se parecen porque cada uno es un ____ del precio.

I find the discount first because ____.
Primero hallo el descuento porque ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A test has 25 questions. A student gets 88% correct. How many are correct?

A. 22

B. 20

C. 23

D. 21

Show your work:

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2. A shirt is $30 after a 25% discount off the original price. What was the original price?

A. $40

B. $37.50

C. $45

D. $36

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A store has a 20% off sale, then takes an additional 10% off the sale price. Is this the same as 30% off the original? Use $100 as the original price. Show your work for both scenarios and explain the difference.

Sentence starter: Taking 20% off then 10% off gives a final price of $____, but 30% off the original gives $____. The difference is $____ because ____.

Show your work:

____

____

____

____

Reflect — Exit Ticket

A jacket costs $80 and is 35% off. Sales tax is 5%. What is the final price?

A. $54.60

B. $52.00

C. $56.00

D. $49.40

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **We Try (notes):** A. \$3.00 *— 15% of \$20 = 0.15 × 20 = \$3.00.*
2. **You Try (notes):** A. \$42.00 *— Tax: 0.05 × \$40 = \$2.00. Total: \$40 + \$2 = \$42.00.*
3. **Try It 1:** A. 22 *— 0.88 × 25 = 22 questions.*
4. **Try It 2:** A. \$40 *— \$30 is 75% of the original; 30 ÷ 0.75 = \$40.*
5. **Exit Ticket:** A. \$54.60 *— Discount: 0.35 × \$80 = \$28. Sale price: \$80 − \$28 = \$52. Tax: 0.05 × \$52 = \$2.60. Final: \$52 + \$2.60 = \$54.60.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Percent is a way to compare a number to 100, shown with the % sign.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).